CGA: Conditioning as Group Action

Richard Bao
richardbao419@gmail.com

Abstract

We present **CGA** (Conditioning as Group Action), a framework that treats feature conditioning $y = T(c)\,x + \beta(c)$ as a group representation of the condition space: the condition c enters the operator *linearly in an abelian Lie algebra*, $T(c) = P\exp(\mathrm{A}(Wc))P^\top$, so the homomorphism law $T(c_1{+}c_2) = T(c_1)T(c_2)$ holds exactly, for any weights, by construction. Choosing the algebra recovers deployed mechanisms as special cases: the diagonal algebra yields an exactly-compositional variant of FiLM (FiLM's usual MLP head parametrizes the same orbit *without* the homomorphism, which we show is precisely what fails under composition); the maximal torus $\mathfrak{so}(2)^{d/2}$ with scalar $c{=}n$ recovers RoPE, and with learned generators the multiplicative GRAPE construction, transplanted from attention positions to arbitrary condition vectors in FiLM's role. The instantiation we study — planar rotations with angles Wc, conjugated by a Group-and-Shuffle structured orthogonal basis — is exactly orthogonal, identity at $c{=}0$ and at initialization, and costs 1.11× FiLM's conditioning FLOPs (a *dense* learned basis provably exceeds any FiLM-comparable budget; at transformer sites the basis can be dropped entirely, as in RoPE). Under a pre-registered, budget-fair protocol against FiLM, conditional LayerNorm, concatenation, and two input-conditioned hypernetworks (up to 48× our parameters), CGA has the lowest compositional OOD error on all five suites we evaluate — two synthetic operator suites and three image-transformation suites on dSprites and 3D Shapes — with complete seed separation throughout ($p{=}9.1{\times}10^{-5}$, $N{=}10$), including zero-shot extrapolation to compositions longer than any seen in training. Swapping only the linear map for an MLP degrades OOD error by up to $4.3{\times}10^4$; the most expressive baselines generalize *worst*. One suite's pre-registered gate returns a negative on a non-OOD criterion — a 1.46× in-distribution fit penalty on 3D Shapes — which we report as registered: exact orthogonality is a robustness–fit trade-off, quantified.

1 The framework

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM [1] computes $y = \gamma(c) \odot x + \beta(c)$; adaLN in DiT blocks [2] is its normalized form; hypernetworks emit $W(c)$ [3]. All are instances of

$$y = T(c)\,x + \beta(c), \tag{1}$$

differing in the family from which the per-sample operator $T(c)$ is drawn.

Definition 1 (Compositional conditioner)**.** *$T : \mathbb{R}^k \to \mathrm{GL}(d)$ is* compositional *if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1{+}c_2) = T(c_1)\,T(c_2)$ and $T(0) = I$.*

Proposition 1 (Exact composition by construction)**.** *Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $\mathrm{A} : \mathbb{R}^m \to \mathfrak{a}$ linear, $W \in \mathbb{R}^{m\times k}$, and P orthogonal. Then $T(c) = P\exp(\mathrm{A}(Wc))\,P^\top$ is a compositional conditioner for* any W, P*; if $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.*

Proof. $\exp(X)\exp(Y) = \exp(X{+}Y)$ for commuting X, Y; $\mathrm{A}\circ W$ is linear; conjugation preserves products and $\exp(0) = I$. □

Composition is thus a property of the *parametrization* — it holds at initialization, during training, and off-distribution. Everything learnable (W, P) only selects *which* one-parameter subgroups the condition drives and *how fast*.

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms exactly:

Table 1: Compositional OOD error (mean over 10 seeds). CGA is lowest in every column with complete seed separation vs. the strongest hypernetwork baseline (p=9.1×10^{-5}, δ=−1.0). S1–S3b pass the full pre-registered gate; S4 returns a registered negative on a separate in-distribution criterion (text).

Conditioner	S1	S2	S3	S3b	S4
FiLM	1.3×10^{-2}	1.6×10^{-2}	3.0×10^{-3}	5.9×10^{-3}	4.5×10^{-3}
Concat-MLP	1.5×10^{-2}	1.1×10^{-2}	2.9×10^{-3}	6.3×10^{-3}	5.4×10^{-3}
Cond-LN	2.4×10^{-2}	2.7×10^{-2}	3.2×10^{-3}	6.1×10^{-3}	8.6×10^{-3}
Hypernetwork	1.9×10^{-3}	3.4×10^{-3}	3.8×10^{-3}	7.0×10^{-3}	9.3×10^{-3}
Dyn. low-rank	5.7×10^{-3}	6.1×10^{-3}	5.1×10^{-3}	8.1×10^{-3}	2.0×10^{-2}
MLP-head (abl.)	–	1.1×10^{-3}	2.7×10^{-3}	5.0×10^{-3}	3.3×10^{-3}
Lie (ours)	7.4×10^{-4}	3.3×10^{-8}	1.8×10^{-3}	4.0×10^{-3}	1.2×10^{-3}

$\mathfrak{a}$	condition	P	recovered mechanism
$\mathrm{diag}(\mathbb{R}^d)$	c	I	*exponential FiLM*: $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	**this work** (FiLM's role)

Standard FiLM parametrizes the same diagonal orbit through an MLP head $\gamma(c)$ — the orbit is right, the homomorphism is absent, and §2 shows empirically that this is exactly what fails under composition: our ablation with an MLP angle head (same orbit, same basis, more compute) loses by up to 4.3×10^4. The PEFT literature (LoRA, OFT, BOFT, HRA, Group-and-Shuffle [6–10]) owns these operator families as *fixed weight* adaptations; CGA places them in FiLM's role — per-sample, input-conditioned, activation-space — where the composition question exists.

Cost, and where the basis comes from. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, exp is closed-form planar rotation: $3d$ FLOPs, no matrix exponential. The basis is the entire budget question. A dense learned P costs $4d^2$ FLOPs per sample (1.52× FiLM's whole conditioning path at d=128) — infeasible under any FiLM-comparable ceiling. Two resolutions: **(i)** a structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, à la Group-and-Shuffle): total 1.11× FiLM, undercomplete (4,800 parameters vs. $\dim \mathrm{SO}(128)$=8,128) yet sufficient, since P need only expose the planes the condition acts on; **(ii)** at transformer sites, *no explicit* P: the adjacent dense projections absorb the basis, which is exactly how RoPE deploys. The operator satisfies identity-init (W=0 $\Rightarrow T \equiv I$), $T(0) = I$ structurally, and $\sigma(T(c)) \equiv 1$; all are unit-tested invariants of the release.

2 Experiments

Protocol. Pre-registered throughout: margins, tests (N=10 seeds, one-sided Mann–Whitney $p \le 0.01$, Cliff's $|\delta| \ge 0.474$), and single-read OOD splits fixed before each run; equal backbones; a shared counter enforces conditioning cost $\le$ 1.20× FiLM and parameters $\le$ 1.05× the smallest hypernetwork baseline (baselines may be larger — up to 48×). Verdicts are reported as registered, including the negative. Baselines: FiLM, conditional LayerNorm, Concat-MLP, dense hypernetwork I+$\Delta W(c)$, dynamic low-rank I+$A(c)B(c)^\top$; plus the MLP-head ablation (identical basis, more compute, no linearity).

Suites. S1/S2 (synthetic): learn $y = M(c)x$ for multi-hot compositions of 8 rotation primitives; S2 hides the primitives' planes behind a random basis B and additionally tests all 56 never-trained *triples*. **S3/S3b (dSprites) and S4 (3D Shapes):** conditional image transformation — encode a real image, apply $T(\Delta)$ on the learned latent for a factor-change Δ, decode; score pixel MSE against the dataset's unique ground-truth image; OOD = never-trained two-factor change types; S3b/S4 include a *categorical* (shape) factor encoded as a one-hot difference.

Findings. (1) Group-structured conditions: solved, not approximated. On S2, CGA reaches 3.3×10^{-8} OOD MSE — five orders below the best baseline — and 5.3×10^{-8} on never-trained triples: error does not grow with composition

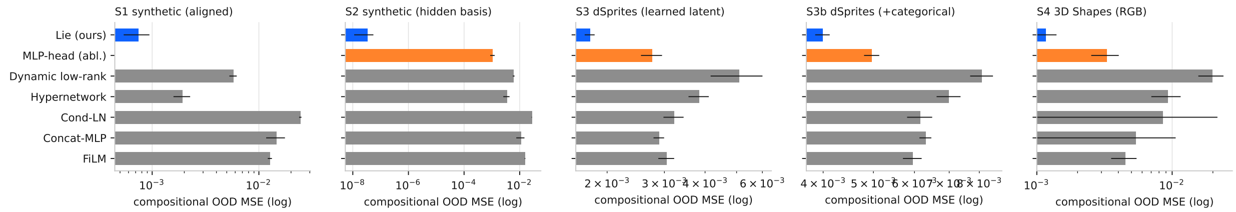

Figure 1: OOD error by conditioner (log scale; blue = CGA, orange = MLP-head ablation).

length. Learned angles recover the true generators to $2{\times}10^{-4}$ rad; a coordinate-aligned ablation ($P{=}I$) collapses to FiLM's error, so basis *discovery* carries the result. **(2) The advantage survives learned latents.** On dSprites all arms fit in-distribution identically, isolating composition: CGA's OOD/in-dist gap is 1.25× (1.63× with the categorical factor) vs. 2.1–3.3× for every baseline, a 53.8% (42.9%) margin over the strongest baseline at 39× fewer conditioning FLOPs. **(3) Linearity is the mechanism.** The MLP-head ablation — same basis, same orbit, more compute — is $4.3{\times}10^4$× worse on synthetic triples and 1.55× worse on dSprites. **(4) Capacity inverts.** The two hypernetwork arms, the only ones able to represent *any* operator, have the worst compositional gaps on images (2.7–26×): expressivity is spent memorizing training condition types. **(5) The registered negative.** On 3D Shapes, CGA again has the best OOD error (87.5% margin; gap 1.13× vs. 13–26×) but pays a 1.46× in-distribution fit penalty that trips the pre-registered no-regression gate ($\leq$ 1.10×). We report the suite as a negative, as registered: exact orthogonality is a robustness–fit trade-off, and our protocol prices it instead of hiding it.

3 Discussion

CGA does for conditioning what RoPE/GRAPE did for position: it moves a desirable law — composition — from something a network must learn into something the parametrization guarantees. The evidence here is deliberately controlled (64×64, known factors); the deployment experiment is the adaLN site of diffusion transformers, where the basis-free variant drops in at below-FiLM marginal cost. Two boundaries are already quantified: categorical factors attenuate (but do not remove) the advantage, and richer scenes can price the orthogonality constraint in in-distribution fit — motivating the abelian-algebra generalizations of Proposition 1 beyond the pure torus (e.g., torus × bounded diagonal) as the next design point. Adjacent prior work learns group actions on latents for world models [11, 12]; CGA differs in living in FiLM's role under budget parity, and in that composition is exact by construction rather than a training objective.

Reproducibility. All specs pre-registered; every number regenerated from append-only logs; operator invariants and fairness counters are unit tests; amendments and one accounting erratum disclosed in the repository.

References

[1] E. Perez et al. FiLM: Visual reasoning with a general conditioning layer. *AAAI*, 2018.

[2] W. Peebles, S. Xie. Scalable diffusion models with transformers. *ICCV*, 2023.

[3] D. Ha, A. Dai, Q. Le. Hypernetworks. *ICLR*, 2017.

[4] J. Su et al. RoFormer: Enhanced transformer with rotary position embedding. *arXiv:2104.09864*, 2021.

[5] Anonymous. Group representational position encoding. *ICLR*, 2026. arXiv:2512.07805.

[6] E. Hu et al. LoRA. *ICLR*, 2022.

[7] Z. Qiu et al. Orthogonal finetuning. *NeurIPS*, 2023.

[8] W. Liu et al. BOFT. *ICLR*, 2024.

[9] S. Yuan et al. Householder reflection adaptation. *NeurIPS*, 2024.

[10] M. Gorbunov et al. Group and shuffle: Efficient structured orthogonal parametrization. *NeurIPS*, 2024.

[11] R. Quessard, T. Barrett, W. Clements. Learning disentangled representations and group structure of dynamical environments. *NeurIPS*, 2020.

[12] H. Caselles-Dupré, M. Garcia-Ortiz, D. Filliat. Symmetry-based disentangled representation learning requires interaction with environments. *NeurIPS*, 2019.